

5/12/2022

Computer Architecture:-

First μp - Intel Intel 4004

ARM type architecture

X86 type architecture.

Speed, throughput, latency

RISC

CISC

Bus: $\rightarrow$ Address bus

$\rightarrow$ Data bus

$\rightarrow$ Control bus

8085 $\mu p \rightarrow$ 8 bit microprocessor i.e, it can carry 8 bit information at a time $\rightarrow$ external memory - 64KB.

Programming model of 8085

Accumulator (8-bits)	Flags (8-bits)
B (8)	C (8)
D (8)	E (8)
H (8)	L (8)
Program Counter PC (16 bit)	
Stack pointer SP (16)	

2 0 0 1
0010 0000 0000 0001

01001 1001
10001 0000
11010 1001
17 15 14 13 12 11 10 9
1 2 3 4 5 6 7 8
- A 9

$= 2^8 + 2^7 + 0 + 8 + 0 + 0 + 1$
 $= 169$

1001 $\rightarrow$ 9
1010 $\rightarrow$ 10 A

1011 $\rightarrow$ 11 B

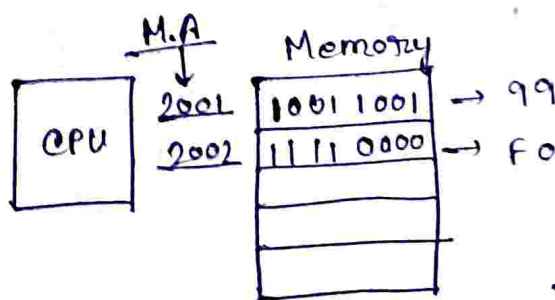
1100 $\rightarrow$ 12 C

1101 $\rightarrow$ 13 D

1110 $\rightarrow$ 14 E

1111 $\rightarrow$ 15 F

4 2 1
1 0 1 0 A 10
1 0 1 1 B 11
1 1 0 0 C 12



1 1 0 1 13 D
14

2 0 0 1
9 9
15 0
6
1001 1001
1111 0000
000 11000 1001
1 8 9

$$\begin{array}{cccccccc} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

$$128 + 0 + 0 + 16 + 8 + 1$$

$$= 153$$

$$1 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0$$

$$128 + 64 + 32 + 16 = 240$$

$$\begin{array}{r} 240 \\ -153 \\ \hline +87 \end{array}$$

$$\begin{array}{ccccccc} 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ \hline & 2 & & 3 & & 1 & & & \end{array}$$

$$\begin{array}{ccccccc} 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ \hline & 3 & & 6 & & 0 & & & \end{array}$$

$$\begin{array}{ccccccc} 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ \hline & 1 & & 1 & & 4 & & 4 & & & & \end{array}$$

$$(1 \ 1 \ 0 \ 0 \ . \ 0 \ 0 \ 1 \ 1)_2$$

$$\rightarrow (C \ . \ 3)_{16}$$

$$2^7 \ 2^6 \ 2^5 \ 2^4 + 2^3 \ 2^2 \ 2^1 \ 2^0$$

$$8 + 4 + 0 + 0 + 0 + 0 + \frac{1}{8} + \frac{1}{16}$$

$$= \frac{128 + 64 + 2 + 1}{16}$$

$$= \frac{195}{16}$$

$$(0 \ 0 \ 1 \ 1 \ . \ 1 \ 0 \ 0 \ 1)_2$$

$$\rightarrow (3 \ . \ 9)_{16}$$

$$2^3 + 2^2 + 2^1 + 2^0 + 2^{-1} + 2^{-2} + 2^{-3} + 2^{-4}$$

$$= 0 + 0 + 2 + 1 + \frac{1}{2} + 0 + 0 + \frac{1}{16}$$

$$= \frac{32 + 8 + 1}{16}$$

$$= \frac{41}{16}$$

$$\frac{195}{16} - \frac{41}{16} = \frac{154}{16} = 9.625$$

$$\frac{195}{16} + \frac{41}{16} = \frac{236}{16} = 14.75$$

- ① Digital logic ———→ ① Boolean → Gates → Transistors → Semi-Conductor
② Microprocessors ———→ ② Combinational circuits (Morris Mano)
③ CA ———→ ③ Sequential circuits.

Universal Gates:-

NAND, NOR, NOT

4

transistors

2

transistors (silicon)

NAND + NOT = AND

NOR + NOT = OR

Digital Logic

Combinational Circuits

Sequential Circuits

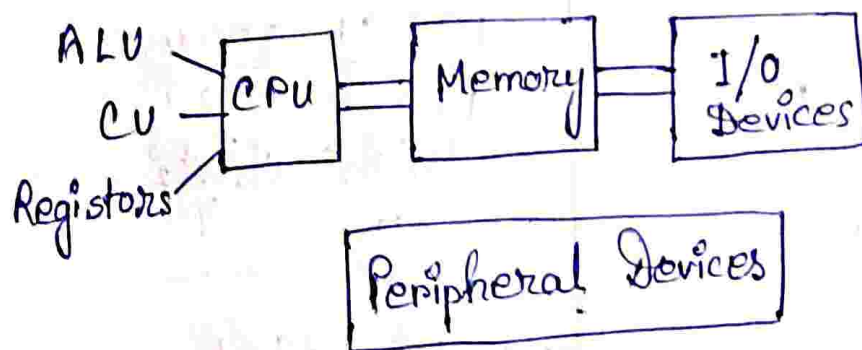
Microprocessor

INTEL-4004

1975 — 8085 μP
8086 μP

4 bit → 0001

8 bit → 0000 1111



Low Level Language - Assembly Language

High Level language - C, Java

08/12/2022

Computer Architecture:-

- ① First ever microprocessor is the Intel 4004, a 4-bit CPU by Intel Corporation in 1971.
- ② Silicon based transistors

Distribution of Semester:-

- 1) Minor - 15%
- 2) Midterm - 20% - 25%
- 3) Final - 50%
- 4) Project - 15% on Assembly Language (Optional)

- application purpose - projector, smart watch
- Home - 220 - 250 Volt Volt
50-60 Hz.
- Every electronic appliances work on low voltage DC.

③ 8085 μP

Major features:-

- 1) 8 bit μP general purpose because μP .
(laptop, desktops)

- 2) 64 KB of memory $\rightarrow$ Extended memory.  40 pins.

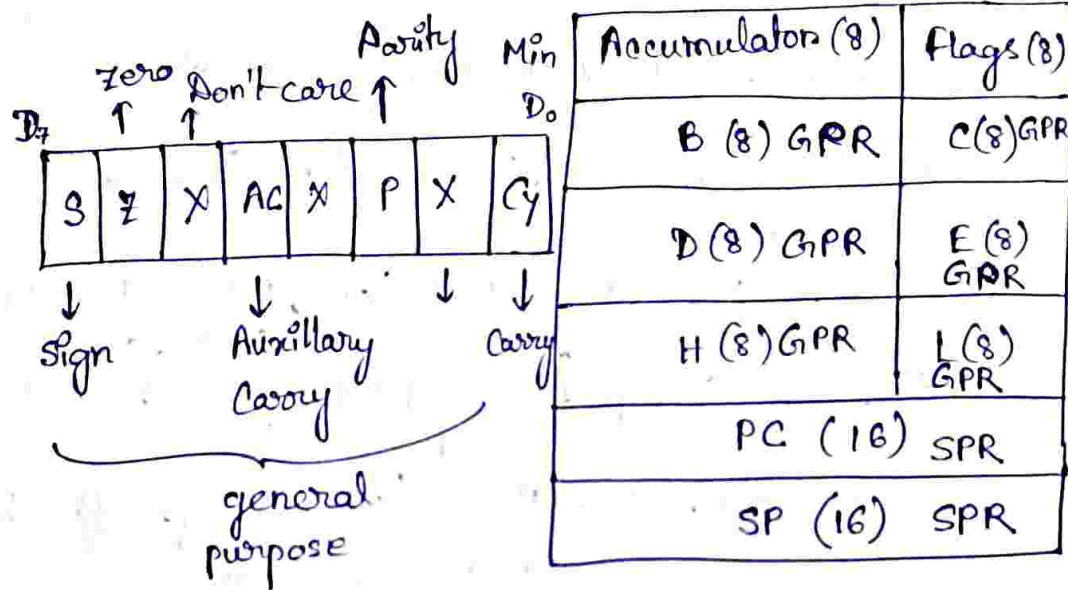
- 3) 5V_{DD} power supply [0 to 5 voltage]

- 4) Operation frequency - 3 MHz (Clock)

- 5) Buses: Address $\rightarrow$ Unidirectional, helps to identify peripheral,
DataControl $\rightarrow$ Bidirectional, transfers data
ControlData $\rightarrow$ Uni, synchronisation of overall computer system.

memory alters

No. of buses is 8 as 8-bit processor.



Special
SPR - Specific Purpose Registers

GPR - General Purpose Registers

Application Specific processor
e.g. Lift, Automobile etc, projector.

- A program counter is a 16 bit register that is used to control the sequencing of execution of an instruction.

It holds the address of next instruction.

(because it points to 16 bit address)

- SP - A 16 bit register that is used to point into the stack or in other words memory. The stack is an area of memory used to hold the data that is required to be retrieved soon. (similar to cache memory)

The stack is used to access the memory in a LIFO fashion. [LIFO - Last In First Out]

PUSH } operations.
POP }

Metro card → Magnetic Storage.

• Instructions can be of 5 different sub-groups

$$\overline{P+Q} \quad P+Q \\ = \overline{P+Q} + \overline{P+Q}$$

1) Data transfer Operation e.g. Mov, Mov immediate, etc.

2) Arithmetic Operations e.g. Add, Subtract, Add Imm, Sub Imm.

3) Logical Operations e.g. XOR, AND, NAND, etc, RLC, RAL, RRC, RAR, compare, *

4) Branching Operations e.g. Jump (for jumping addresses within program)

5) Machine Control operations e.g. HLT, (At end for finishing program)
NOP

~~RAL~~ ~~RAR~~ - Rotate Accumulator Left involving Cy flag in rotation.

RAR - Rotate Accumulator right (through the carry) involving Cy flag in rotation

RRC - Rotate Right Accumulator. * compare immediate.

RLRAL - Rotate Left Accumulator

JP - (Jump if Positive), Ma.

JMP JM - (Jump if Minus)

Conditional Branching - JZ - (Jump Zero), JC - (Jump if Carry)

Unconditional Branching - JNZ - (Jump if not Zero), JNC - (Jump if not carry)

- Call " " " "

- Return " " " "

HLT - Stops executing the program.

NOP - No operation

3E in hexadecimal
 • MVI A, 32 - Move immediate into Accumulator
 OP code operand
 32 is the data in hexadecimal.

3 E 3 2
 0011 1110 0011 0010

JMP 2001
 C3 20 01

1100 0011 0010 0000 0000 0001 0001

A-10

B-11

C-12

D-13

E-14

F-15

2 | 14, 0

2 | 7, 1

2 | 3, 1

1

2 | 12, 0

2 | 6, 0

2 | 3, 1

1

2 | 20, 0

2 | 10, 0

2 | 5, 1

2 | 2, 0

1

• Addressing Mode:- A The μP has different modes of specifying the data for the instruction, which are called the addressing modes. A different types of addressing modes for 8085 μP .

① Implied - CMA

② Immediate

③ Direct

④ Indirect - LDAX B

⑤ LDA → Load into the Accumulator (LDA 2001)

Computer Architecture:-

1) Combinational Circuits:-

Multiplexers → De Multiplexers

Encoder → Decoder

HA → FA, FA

IBM, Western Universities,
TJ Watson, NPTEL lectures 12/12/2022
Samsung, Google, Intel
Apple,

2) Sequential Circuits

Latches

Flip flop

Registers

Counters

3) Processors Performance:- Parameters

① Speed

④ Through put

② Latency

⑤ Bandwidth

③ Power consumed

1990 - GUN

No mouse during
8085 μ P.

④ Advanced version of 8086 is x86.

⑤ 8086 is a type of cisc architecture.

⑥ Two major type architecture:

① ARM architecture

8086

8087

186

286

386

486

Pentium

F F F F
1111 1111 1111 1111

1000 8

1001 9

1010 A

1011 B

1100 C

1101 D

1110 E

1111 F

⑦ 8086 Microprocessor:-

→ Important pinouts

→ 40 pins

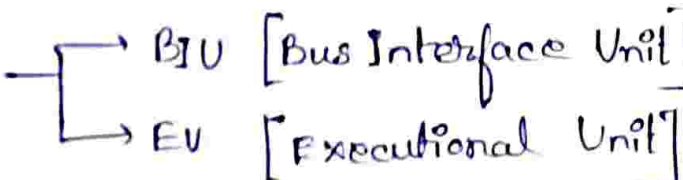
→ Features:-

- 16-bit ALU

- 16-bit data bus (8088 has 8-bit data bus)

- 20-bit address bus - $2^{20} = 1,048,576 = 1 \text{ meg}$

- The add. refers to a byte in memory.

→ 8086 has two parts — 

→ Memory 1MB

→ BIU — fetches instructions, reads and writes data and computes the 20-bit address.

→ EU — decodes and executes the instructions using the 16-bit ALU.

→ Fetch → Decode → Execute → Store

AH	AL
BH	BL
CH	CL
DH	DL
SP (16)	
BP (16)	
SI (16)	
DI (16)	

SI = Source index

DI = Destination index

SP = Stack Pointer

BP = Base pointer

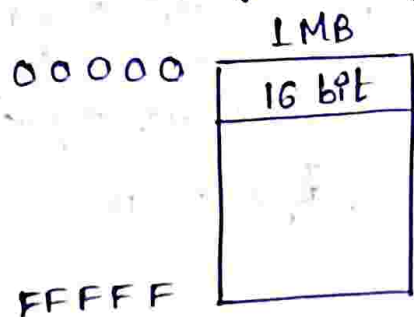
AH = A higher order

AL = A Lower order .

15/12/22 Computer Architecture:-

Low level programming language - Assembly

High Level programming language - C++, C



8086 μ p

20 bit Address bus

Accumulator - 16 bit

BCD $\rightarrow$ 16 bit

Programming model of 8086:

① BEV $\rightarrow$ use to identify the bus and locate the particular address where your data is stored.

② EV

① Von Neuman Architecture

② Harvard Architecture

* Sequential / Combinational circuit

* RISC / CISC

Computer Architecture:-

21/12/2022

8086 Programming Model:-

① BIU register: (20 bit address)	ES	Extra Segment
	CS	Code Segment
	SS	Stack Segment
	DS	Data Segment
	IP	Instruction Pointer.

② EU register: (16 bit arithmetic)	AH(8)	AL(8)
	BH(8)	BL(8)
	CH(8)	CL(8)
	DH(8)	DL(8)
	SP(16)	
	BP(16)	
	SI(16)	
	DI(16)	
	FLAGS(16)	

EU contains the following 16-bit registers:

AX - The Accumulator

BX - The Base register

CX - The Count Register

DX - the Data Register

SP - the Stack Pointer
BP - the Base Pointer } default to stack segment

SI - the Source Index Register

DI - the Destination Register.

③ Flag Register:

15															0
—	NT	IOPL	OF	DF	IF	TF	SF	ZF	—	AF	—	PF	—	CF	

Control Flags:

IF : Interrupt enable Flag

DF : Direction flag

TF : Trap flag.

Status flags:-

CF: Carry Flag

PF: Parity Flag

AF: Auxiliary Carry Flag

ZF: Zero Flag

SF: Sign Flag

OF: Overflow Flag

NT: Nested Task Flag

IOPL: Input/output privilege Level

$$f \propto \frac{1}{T} \quad 3-5 \text{ Hz}$$

$$3 = \frac{1}{T} \Rightarrow T = \frac{1}{3} = 0.$$

Here T denotes time for executing an instruction.

Operations in μP : fetch, decode, execute, store.

Internal architecture of Intel 8085A:

It consists of five essential blocks:

1) ALU Arithmetic Logic section

2) Register Section

3) The interrupt Control Section

4) Serial I/O Section

5) Timing and Control Unit/section

Instruction cycle
Machine cycle
T-states

Control signal inputs - READY, HOLD,

Interrupt control signals - TRAP,
RST7.5, RST6.5, RST5.5 and INTR.

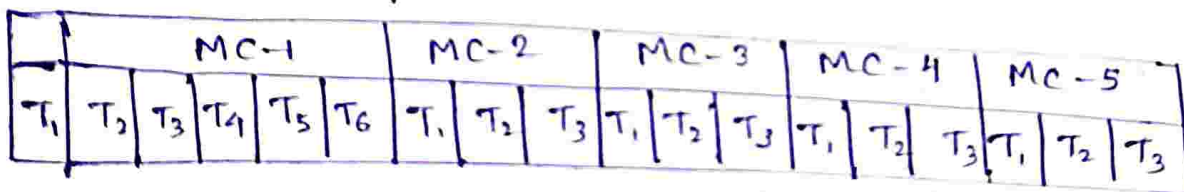
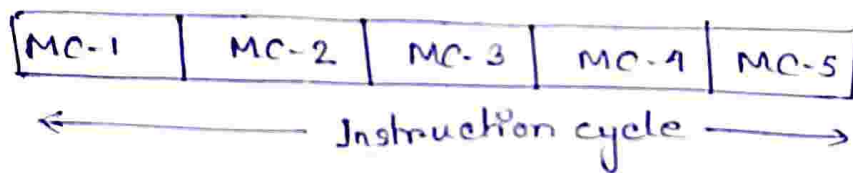
$T_1, T_2, T_3, T_4, T_5, T_6,$

$T_{RESET}, T_{HALT}, T_{WAIT},$

$T_{HOLD}.$

There are 7 type of Machine cycles:

- 1) OFMC - Opcode Fetch Machine Cycle
- 2) MRMC - Memory Read Machine Cycle.
- 3) MWPMC - Memory Write Machine Cycle.
- 4) IORDMC - Input/Output Read Machine cycle
- 5) IOPWPMC - I/O write Machine Cycle
- 6) INTAMC - Interrupt Acknowledge Machine Cycle.
- 7) BIMC - Bus Idle Machine Cycle.



One clock period i.e. the period b/w two negative going transitions of that clock is called T state.

OP CODE - FETCH: 4-6

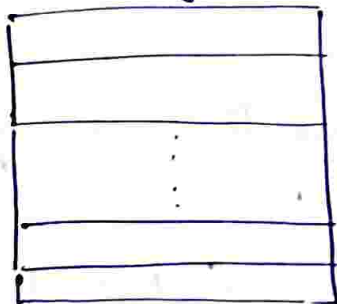
The first 3 states of the T-states are identical to MRMC.
The additional T-states in the OFMC are the T-states required by

Memory Bank

STA $\frac{2020}{8 \ 8 \ 8}$ bits

• Digital Logic Morris Mano.

2001



2020

Mnemonic

Instruction byte

Machine Cycle

STA

OP Code

OFMC

Lower Address

MRMC

Higher Address

MRMC

MWRMC

Thus 3-byte STA instruction has four machine cycles in its instruction cycles:

Instruction Cycle and Machine Cycle for STA ADDR instruction

• Computer Organization:

• Computer Architecture:

→ Performance { Processor
Memory
I/O
Peripherals }

→ Cost ↓

→ Features ↑

→ Size ↓

Performance + Cost + Features + Size

• Moope's Law:-

• Control Unit:-

Computer Architecture:-

22/12/2022

• Inside the CPU (Processor)

→ CU

→ ALU

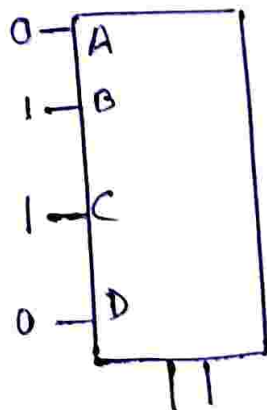
• Contains several registers (GPR and SPR) for temporary storage.

• Logic operation circuit - AND, OR, NOT, Shift, compare, etc.

• Arithmetic operation circuit - +, -, x, ÷, etc. (combinational circuit)

• During instruction execution, the data (operands) are brought in and stored in some registers, → desired operation carried out, and → result stored back in some register/memory.

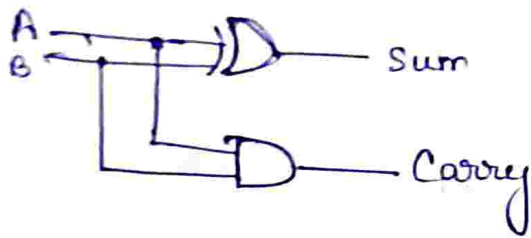
Mux



E	F	O/p
0	0	A = 0
0	1	B = 1
1	0	C = 1
1	1	D = 0

$$X = EF + FE \quad (= \text{XOR})$$

Half Adder

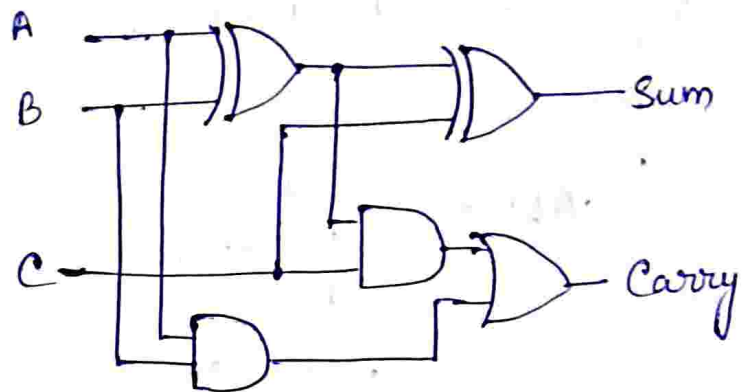


$$\text{Sum} = A \oplus B$$

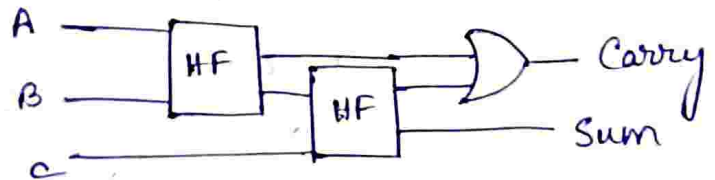
$$\text{Carry} = A \text{ AND } B$$

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Full Adder



||



Role of CU:- ^{senses and} Sends control signals to control states ~~and se~~ of various functional units.

Memory unit:- Primary - RAM
Secondary - HDD

Various types of memory - RAM, ROM, Magnetic Disk, Flash Memory.

→ RAM - which is used for the cache and primary memory sub-system. Read and Write Access times are independent of the location being accessed.

→ ROM - which is used as part of the primary memory to store some fixed data that cannot be engaged.

→ Magnetic Disk - which uses direction of Magnetisation of tiny magnetic particles on a metallic surface to store data. Access times vary depending on the location being accessed, and is used as secondary memory.

→ Flash memory - which is replacing magnetic disks as secondary memory devices. They are faster but smaller in size as compared to disk.

Input Units - Keyboard, mouse, Joystick, Camera.

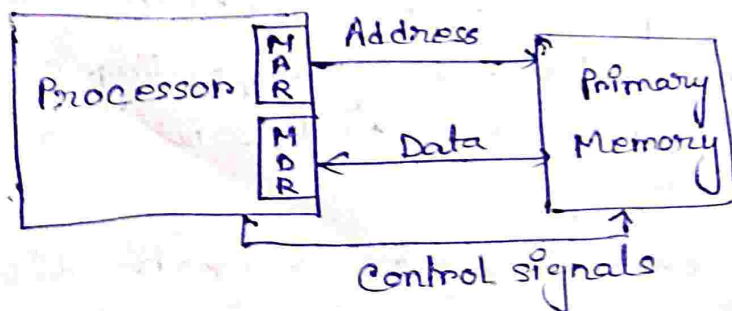
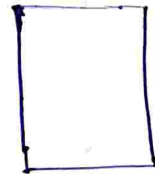
• For interfacing with the Primary Memory:-

→ Two SPR:

→ Memory Address Register (MAR): Holds the address of the memory location to be accessed.

→ Memory Data Register (MDR): Holds the data that is being written into memory or will receive the data being read out from memory.

→ Memory considered as a linear array of storage locations (bytes or words) each with unique address.



• Multi-Bus Architectures:-

- Modern processors have multiple buses that connect the registers and other functional units.
 - Allows multiple data transfer micro-operation to be executed in the same clock cycle.
 - Results in overall faster instruction execution.
- Also advantageous to have multiple shorter buses rather than a single long bus.
 - Smaller parasitic capacitance and hence smaller delay.

- A memory system with M locations and N bits ~~are~~ per location is referred to as $M \times N$ memory.
 - Both M and N are typically some power of 2.

• Some terminologies

• How is Memory Organized.

Half word (2 bytes), Word (4 bytes), Long Word (8 bytes)

For higher data transfer rate, memory is organized such that multiple bytes can be read or written simultaneously.

$$1 \text{ GB} = 8 \times 10^9 =$$

$$1 \text{ TB} = 8 \times 10^{12} =$$

- If there is n bit in the address, the maximum no. of storage locations = 2^n .

530 —

Types of Programs:-

① Application software $\rightarrow$

② System software

Computer Architecture:-

29/12/22.

Digital logic / Microprocessor. } Advance Computer Architecture.

- Distributed Computing
- Edge computing
- Cloud Computing.

Memory systems:- M locations and N bits per location $M \times N$.

- The two conventions while naming a memory is as follows:

① Little Endian — Here the lsb is stored at the lower address which is followed by the msb.

② Big Endian — The msb being stored at the lower address followed by the lsb.

32 bit number

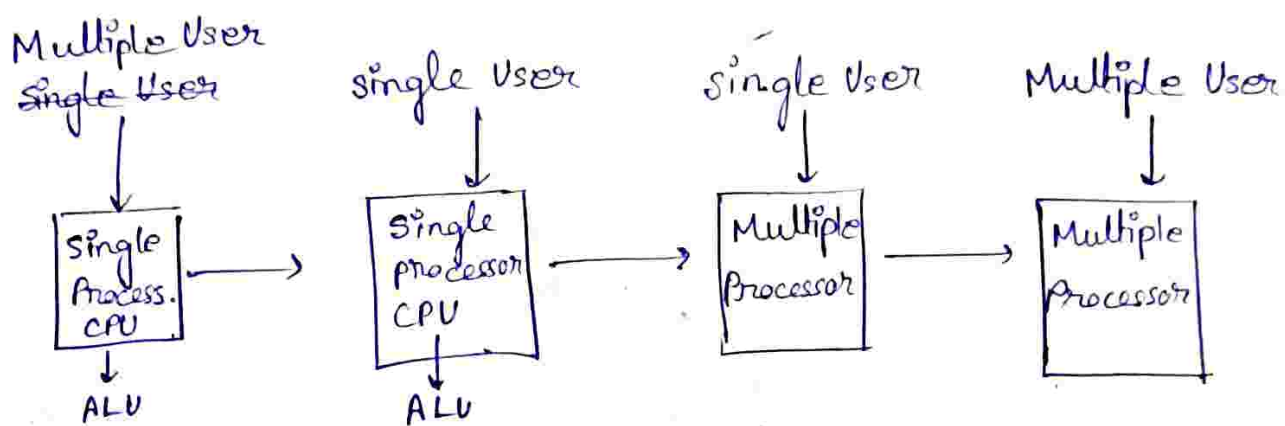
0110 1011 10110101 00110001 0010 1011

<u>Address :</u>	<u>L.E</u>	<u>B.E</u>
2000	00101011	01101011
2001		
2004		

① Little Endian

② Big Endian

- Memory Access Instructions:- Computational speed/Processor speed
Faster Processing $\longleftrightarrow$ Accessing of data/Instructions:
Memory, R/W operations speed,
Peripheral, I/O.



• Assemblers and Compilers:-

Assembler translates the assembly language into machine language (0 and 1).

Compiler translates the high level language into assembly or machine language.

- No. of transistors / No. of Unit area $\rightarrow$ No. of logic gates $\uparrow$
 $\rightarrow$ Address.

No. of Instructions $\uparrow$ (increasing exponentially with time).

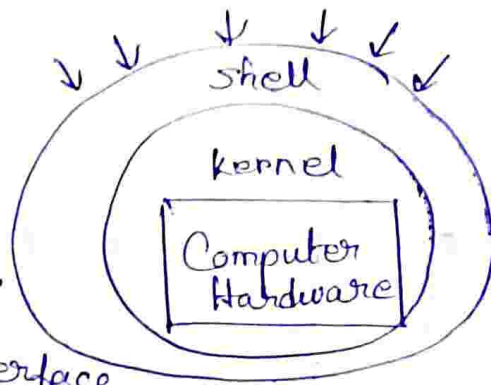
• Software:- If consists of a set of instructions required to solve a (Program) specific problem.

- Types:-
 - Application software - It helps the user to solve user level problem and it may need a system software for its execution.
 - System software - The collection of programs that helps the user to create, analyse and run their program.

• Operating System:-

→ Kernel contains low-level routines for resource management.

→ Shell provides an interface for the users to interact with the computer hardware through the kernel.



• Processors, files, I/O, peripherals are the typical resources for computer system and the tasks performed by the OS as a unit includes: Page 63/61)

→ Assigning memory and disks space to program and data files

→ Moving data b/w I/O devices, memory and disk units.

→ Handling I/O operations, with parallel operations where possible

→ Handling multiple and $\neq$ user programs that are running at the same time.

• Goal of OS: → Classical multi-programming systems.

→ where several user programs loaded in memory

• Switch to another program with when one program gets blocked due to I/O.

• Objective is to maximize resource utilization.

→ Modern time-sharing systems:-
• Widely

Minor-15 (19/01/2023) Till the last Monday (Assembly language)
Mid Term -20 Basic organization
Project/Assignment - 15
Finals: 50

→ Real-time systems. (RTOS)

• specific deadlines:

• Interrupt driven operations — The processor gets interrupted when a new task arrives.
e.g. missile control system, etc.

→ Mobile (phone) systems:

• Two major type of Computer Architecture:-
Von-Neumann and Harvard.

• Pipeline in Executing Instructions:-

→ Instruction execution is typically divided into 5 stages.

• IF • ID • EX • MEM • WB.

Instruction Fetch

Instruction Decode

ALU operations

Memory Access.

Write back result to register file.

Basic 5-stage Pipelining Diagram:-

Instruction	Pipeline Stage						
	IF	ID	EX	MEM	WB		
1	IF	ID	EX	MEM	WB		
2		IF	ID	EX	MEM	WB	
3			IF	ID	EX	MEM	WB
4				IF	ID	EX	MEM
5					IF	ID	EX
Clock Cycle	1	2	3	4	5	6	7

Computer Architecture:-

16/01/2023

X86 configuration is an extended version of 8086 μp .
CISC Architecture.

- T-state
4-6
- Machine cycles
4-6
- Instruction cycle

CISC Architecture/RISC Architectures

Two broad classification of Instruction Set Architecture.

- ① CISC - Complex instruction set Computer.
- ② RISC - Reduced instruction set Computer.

Features of CISC architecture:

- ① It is traditional
- ② Complex in the instruction set architecture.
- ③ Large no. of addressing modes.
- ④ It has special purpose registers and flags.
- ⑤ It has variable length instruction.

- ⑦ Ease of mapping high level language to your machine instruction.
- ⑧ Instruction decoding / complex design of CU.
- ⑨ Pipeline implementation becomes quite complex.

examples:- IBM 360/370

VAX-11/780

Intel x86/Pentium

Features of RISC Architecture:-

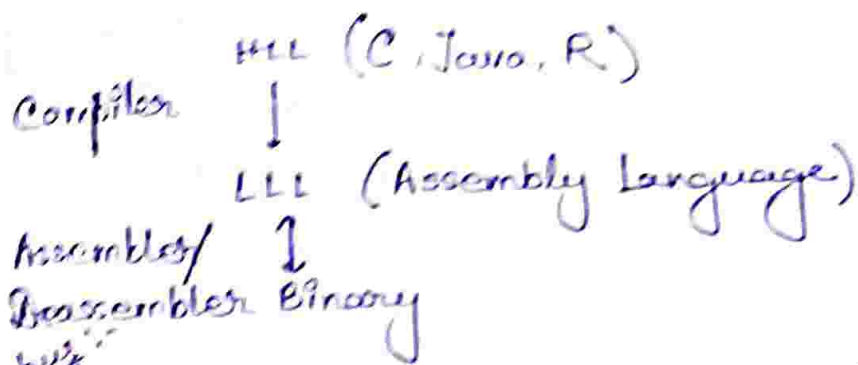
- ① Also referred as load-store Architecture.
- ② Only LOAD and STORE instructions have access to the memory.
- ③ All other instructions operate on the processor registers.
- ④ Simple in terms of pipelining process.
- ⑤ It is simple instruction set in terms of having very few addressing modes.
- ⑥ It has large no. of general purpose registers and small no. of very few special purpose registers.
- ⑦ Instruction length and encoding uniform for easy ~~interaction~~ instruction decoding.
- ⑧ Compiler assisted scheduling of pipeline for improved performance.

examples:- CISC 6500

MIPS family

SPARC

ARM microcontroller family.



Substans
Von Neumann, Howard, 8085/8086 μ P, introduction Pipeline,
Assembly Language. (Addressing modes)

Pattern
4-53 questions $\rightarrow$ 5+5+5

Difference b/w RISC and CISC architectures:-

CISC

- INTEL, VAX, IBM
- The RISC had many times better performance than CISC.
- Less hardware was required for RISC than compared to CISC.

RISC

- ARM, POWER PC, MIPS, M2000
- The RISC required execution of more number of instruction as compared to CISC.
- Cycles per instruction (CPI) for VAX was about six times larger than that of MIPS.
- Hence, MIPS had three times the performance of VAX.

CISC $\rightarrow$ x86 $\rightarrow$ • It has huge no. of installed base, and backward compatibility of machine code which is important in terms of commercial point of view.
↓ converting into
RISC

- They have adopted a balanced view in terms of CISC instruction set where the hardware translates every CISC instruction into an equivalent RISC instruction internally.

- Basic difference b/w CISC and RISC is
- Why CISC is still in use

Computer Architecture: (Week 3)

30/01/2023

CPU-Performance (Most critical part)
Memory $\rightarrow$ RAM

Bhar OS by IIT Madras
based on RISC architecture
-V

• Execution time

- $\rightarrow$ IC (Instruction count)
- $\rightarrow$ CPI (Cycles per instruction)
- $\rightarrow$ Clock cycle (Hz)

- By measuring the execution time one can use to compare two sets of computer say Computer A and a Computer B.
- Performance can be defined as the reciprocal of execution time:

$$\text{Perf}_A = \frac{1}{X T_A} \quad \text{Perf}_B = \frac{1}{X T_B}$$

- We can estimate the speedup of machine A over machine B as:

$$\text{Speedup} = \frac{\text{Perf}_A}{\text{Perf}_B} = \frac{X T_B}{X T_A}$$

e.g. ~~A~~ Three computers A, B, C.

$$A = 2.5 B \quad T_A = 10$$

$$A = 7.5 C \quad T_B = 25$$

$$B = 3.0 C \quad T_C = 75$$

- A program is running on a machine with the following parameters:

— Total no. of instructions executed = 50,000,000

— Average CPI for the program = 2.7

— CPU clock rate = 2.0 GHz (i.e. $C = 0.5 \times 10^{-9}$ sec).

- Execution time of the program:

$$X T = 50,000,000 \times 2.7 \times 0.5 \times 10^{-9} = 0.0675 \text{ sec.}$$

• Factors Affecting Performance:-

	C	CPI	IC
Hardware Technology (VLSI)	X		
Hardware Technology (Organization)	X	X	
Instruction set architecture		X	X
Compiler technology		X	X
Program		X	X

• A trade off:

- RISC : increases no. of instructions/program but decreases CPI and clock cycle time because the instructions are simple and hence the implementations are simple.

- CISC : decreases no. of instructions/program, but increases CPI and clock cycle time because many instructions are more complex.

• Overall, it has been found through experimentation that RISC architecture gives better performance.

• Example. 2 :-

Suppose that a machine A executes a program with an average average CPI of 2.3. Consider another machine B (with the same instruction set and a better compiler) that executes the same program with 20% ^{less} instructions and with a CPI of 1.7 at 1.2 GHz.

What should be the clock rate of A so that the two machines have the same performance.

$$Perf_A = \frac{I}{T_A} = \frac{IC \times CPI \times \frac{10}{12} \times 10^{-9}}{1}$$

$$Perf_B = \frac{I}{T_B} = \frac{(100-20)\% IC \times 1.7 \times \frac{10}{12} \times 10^{-9}}{1}$$

$$\frac{2.3 \times 100}{80 \times 1.7} = 1.6911$$

$$\frac{10}{1.2} = \frac{10}{12} \times 10^{-9}$$

$$f_{\text{clock}} = \frac{1}{T_A} = \frac{1}{IC_A \times 2.7 \times 10^{-9}}$$

$$f_{\text{clock}} = \frac{1}{T_B} = \frac{(100 - 20)\%}{IC_A \times 1.7 \times \frac{10}{12} \times 10^{-9}}$$

$$\therefore f_{\text{clock}} = f_{\text{clock}}$$

$$\Rightarrow 2.3 \times IC_A = \frac{80}{100} \times 1.7 \times \frac{10}{12} \times 10^{-9}$$

$$\Rightarrow IC_A = \frac{80}{100} \times \frac{1.7}{2.3} \times \frac{10}{12} \times 10^{-9}$$

$$= 0.49275 \times 10^{-9}$$

$$= 2.01 \text{ GHz}$$

8 - 1

10^6 - million

10^9 - billion

10^{12} - trillion

• Example-3:

Consider the earlier example with $IC = 50,000,000$; average $CPI = 2.7$ and clock rate = 2.0 GHz.

Suppose we use a new compiler on the same program, for which:

New $IC = 40,000,000$

New $CPI = 3.0$

Also, clock rate = 2.4 GHz.

$$\text{Speed Up} = \frac{T_{\text{old}}}{T_{\text{new}}} = \frac{50,000,000 \times 2.7 \times 0.5 \times 10^{-9}}{40,000,000 \times 3.0 \times \frac{10}{24} \times 10^{-9}}$$

$$= 1.35 \Rightarrow 35\% \text{ faster.}$$

• Instruction types and CPI:

$$\text{CPU clock cycles} = \sum_{i=1}^n (IC_i \times CPI_i)$$

$$\text{Instruction Count (IC)} = \sum_{i=1}^n IC_i$$

IC_i = number of instructions of type i executed.

CPI_i = cycles per instruction for type i .

$$CPI = \frac{\sum_{i=1}^n (IC_i \times CPI_i)}{IC}$$

$$= \sum_{i=1}^n \left(\frac{IC_i}{IC} \times CPI_i \right)$$

• Example-4:

Consider an implementation of a ISA where instructions can be classified into four types, with CPI values of 1, 2, 3 and 4 respectively

Two code sequences have the following instruction counts:

Code-sequence	IC_{type-1}	IC_{type-2}	IC_{type-3}	IC_{type-4}
CS-1	20	15	5	2
CS-2	10	12	10	4

$$\text{Clock cycle (CS-1)} = 20 \times 1 + 15 \times 2 + 5 \times 3 + 2 \times 4 = 20 + 30 + 15 + 8 = 73$$

$$\text{Clock cycle (CS-2)} = 10 \times 1 + 12 \times 2 + 10 \times 3 + 4 \times 4 = 10 + 24 + 30 + 16 = 80$$

Instruction Count.

$$\begin{array}{l|l} \text{CS-1} & \text{CS-2} \\ \hline 20+15+5+2 & 10+12+10+4 \\ = 42 & = 36 \end{array}$$

$$\therefore \text{CPI} = \begin{cases} \text{CS-1} = \frac{73}{42} = 1.738 \\ \text{CS-2} = \frac{80}{36} = 2.222 \end{cases}$$

• Instruction frequency and CPI:-

- F_i denotes the frequency of execution of instruction type i .

$$\text{CPI} = \frac{\sum_{i=1}^n (IC_i \times CPI_i)}{IC}$$

$$F_i = \frac{IC_i}{IC}$$

$$= \sum_{i=1}^n \left(\frac{IC_i}{IC} \times CPI_i \right)$$

$$\text{CPI} = \sum_{i=1}^n (F_i \times CPI_i)$$

• Example-5:-

Instruction type

load

store

ALU

Branch

frequency

20%

8%

60%

12%

CPI
4
3
1
2

$$\text{CPI} = \sum_{i=1}^n (F_i \times CPI_i)$$

$$= \frac{20}{100} \times 4 + \frac{8}{100} \times 3$$

$$+ \frac{60}{100} \times 1 + \frac{12}{100} \times 2$$

$$= \frac{80}{100} + \frac{24}{100} + \frac{60}{100}$$

$$+ \frac{24}{100}$$

$$= \frac{188}{100} = 1.88$$

• Amadahl's Law:-

- Amadahl's law was established in 1967 by Gene Amadahl.
- Basically provides an understanding on scaling, limitations and economics of parallel computing.
- Forms the basis for quantitative principles in computer system design.
- It can be used to find the maximum expected improvement of an overall system when only part of the system is improved.
- It basically states that the performance improvement to be gained from using some faster mode of execution is limited by the fraction of the time the faster mode can be used.

example:- Suppose we are improving a part of the computer system that affects only 25% of the overall task.

- The improvement can be very little or extremely large.
- With 'infinite' speedup, the 25% of the task can be done in 'zero' time.

$$\text{Maximum possible speedup} = \frac{X_{T_{orig}}}{X_{T_{new}}} = \frac{1}{1 - 0.25} = 1.33$$

We can never get a speedup of more than 1.33

- Amadahl's law concerns the speedup achievable from an improvement in computation that affects a fraction F of the computation, where the improvement has a speedup of S .

Before improvement

$1 - F$	F
---------	-----

After improvement

$1 - F$	F/S
---------	-------

• Execution time before improvement: $(1 - F) + F = 1$

• Execution time after improvement: $(1 - F) + F/S$

• Speedup obtained: $\text{Speedup} = \frac{1}{(1 - F) + F/S}$

• As $S \rightarrow \infty$, Speedup $\rightarrow 1/(1 - F)$ obtained.

- The fraction F limits the maximum speedup that can be

Computer Architecture:

23/01/2023

Control bus can be unidirectional or bidirectional depending upon structure of single core to multi-core CPUs.

• Measuring the CPU performance:-

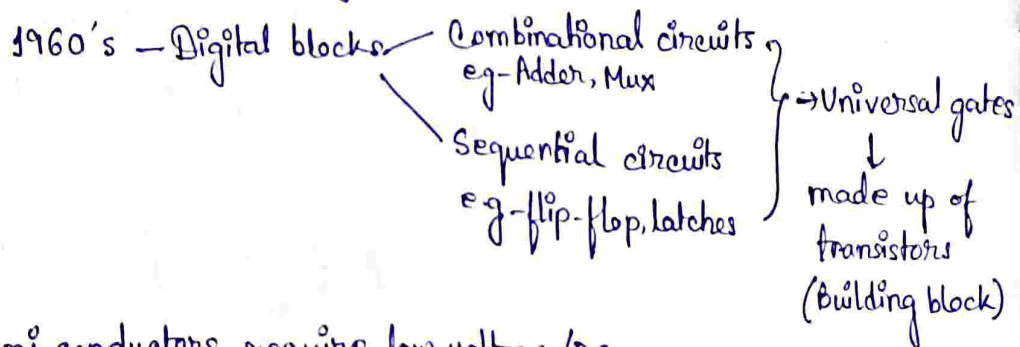
Most processors execute instructions in synchronous way using clock.

$$\text{clock} = \frac{1}{f} \quad f: \text{clock rate.}$$

Clock rate depends upon two factors-

- (i) Technology $\rightarrow$ Transistors
- (ii) CPU Organisation $\rightarrow$ CISC and RISC

A machine instruction consists of no. of elementary microoperations that vary in no. & complexity depending on instruction and CPU organisation used.



Semi-conductors require low voltage/DC.

1970's - 4004 - Transistor of size $10 \times 10^{-6} \text{ m} \rightarrow 10 \times 10^{-9} \text{ m}$ (reached today)

• Minimization of transistors:-

- (i) Package density: Minimization $\uparrow$ density
 - (ii) Computational power: Minimization increases (C.P)
 - (iii) Power requirements: Minimization $\downarrow$ (P.R)
- VGA $\rightarrow$ Visual Graphic Array
Specific purpose.

Any kind of operation in CPU system or H/W that can be carried out in one clock cycle. A single machine instruction may take one or more CPU cycle to complete.

Characterise Instruction:-

Imp { CPS - Cycle per Instruction
Avg. CPI of a program can be characterized as avg. CPI of all instruction executed in a given processor.

Avg. CPI uses different instruction with different CPI.

Imp For a given program compiled on specific machine, we can define the following parameters-

① Instruction count - Total no. of instruction to be executed.

② CPI - Cycles per instruction.

③ Clock cycle (Hz) - (also related to computational power - Clock).

Execution Time: $IC * CPI * CLK$.

Graph Theory:-

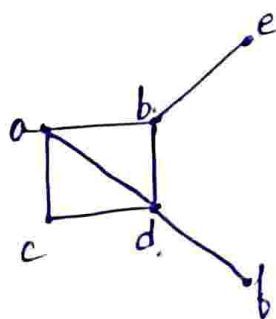
• The no. of vertices of odd degrees in a graph is even.

$$\sum_{i=1}^n d(v_i) = 2e = \sum_{\text{odd}} d(v_j) + \sum_{\text{even}} d(v_k)$$

• Walk - Alternating sequences of the vertices and edges in which we might repeat the vertices. (f-d-c-a-d-b)

• Path - Vertices will not be repeated in the path. (f-d-a-b)

• Closed path - start and end at the same vertices.
(a-b-d-c-a)



• Connected graph - We can reach any vertex from any vertex.

• Disconnected graph - If we can't reach any vertex from any vertex.

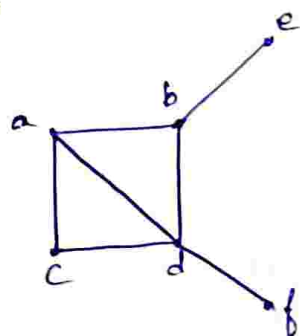
• Length of path - no. of edges in a path.

• Components - A graph is disconnected if it has many components.

• i

i = isolated point.

Example of a disconnected graph.

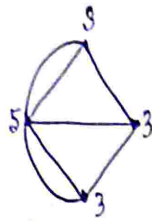


• Let, $G = (V, E)$ is a graph and R is a relation on V which is defined by xRy iff $x = y$ or there is a path from x to y . Then we say, R is an equivalence relation for undirected graph.

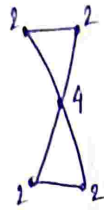
1 component = connected graph.

many components = equivalence classes = disconnected graph.

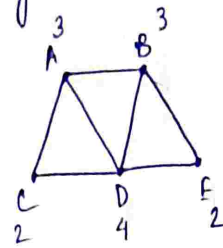
• Euler Graph:- All the vertices of even degree.



Not a Euler graph



Euler graph.



Unicursal graph.

[Open Euler line is, Start point $\neq$ End point]

Book.

Narsingh Deo Graph Theory:

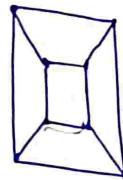


$\approx$

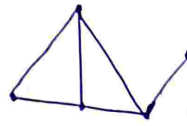
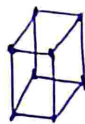


Isomorphic graph.

One-to-one mapping.



$\approx$

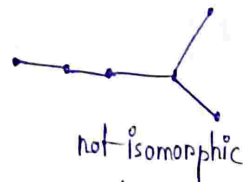


$\approx$

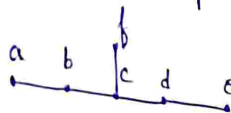


Conditions for isomorphic graph:-

- (i) No. of vertices same
- (ii) No. of edges same
- (iii) No. of degrees of vertices same.
- (iv) Exactly one-to-one correspondence between the vertices.



not isomorphic



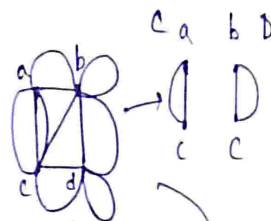
1st Level of Isomorphism:-

• Sub-graphs:- Part of a graph
Sub-set of vertices and the edges connecting those vertices.



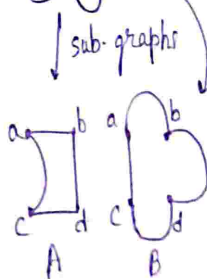
(i) Edge disjoint sub-graph:-

A and B have no edge in common so, they are edge disjoint subgraph.



(ii) Vertex disjoint sub-graph:-

C and D have no vertex in common, so, they are edge vertex disjoint subgraph.

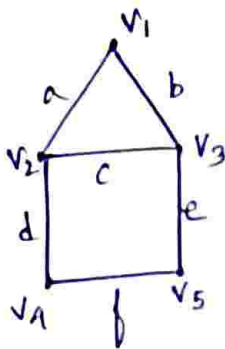


• Vertex disjoint subgraph is always edge disjoint but not the other way around.

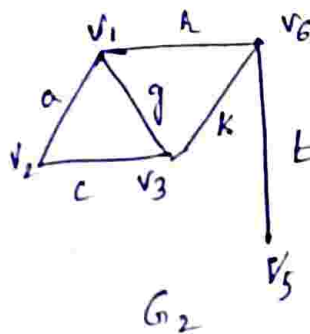
• We define graph using two sets V and E , so set operations are valid on graph such as Union and Intersection.

e.g. ① $G_1 = (V_1, E_1)$ $G_3 = (V_3, E_3) = G_1 \cup G_2$
 $G_2 = (V_2, E_2)$ where $V_3 = V_1 \cup V_2$
 $E_3 = E_1 \cup E_2$

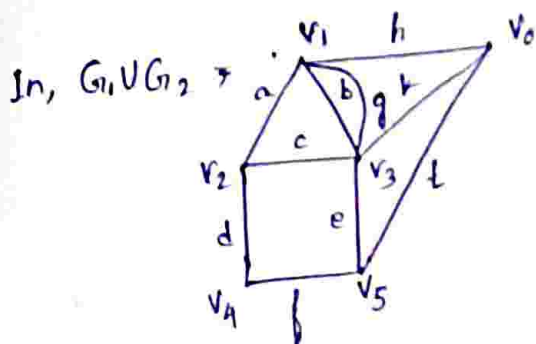
② $G_4 = (V_4, E_4) = G_1 \cap G_2$ where, $V_4 = V_1 \cap V_2$ (common vertices)
 $E_4 = E_1 \cap E_2$ (common edges)



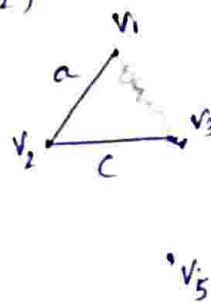
G_1



G_2



In, $G_1 \cap G_2$,

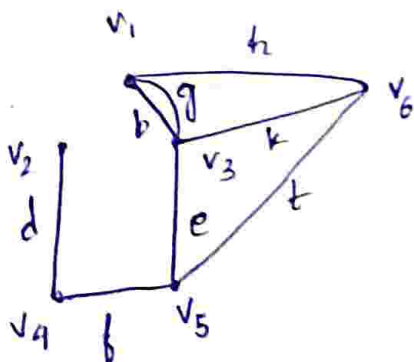


G_3

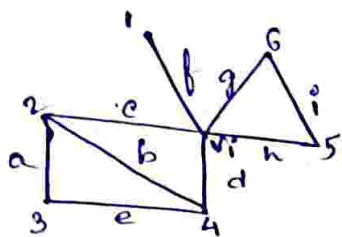
• Ring Sum operation:- $G_5 = G_1 \oplus G_2 = (V_5, E_5)$

$$V_5 = V_1 \cup V_2$$

$E_5 = E$ exclude the common edges.

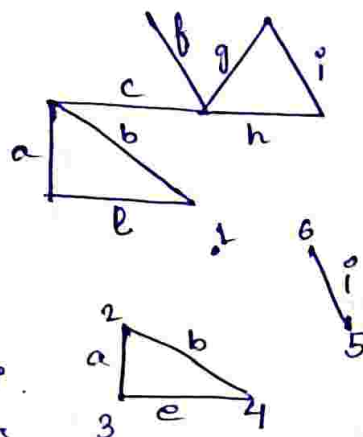


- Det Deletion: (Deletion of edges or vertices).

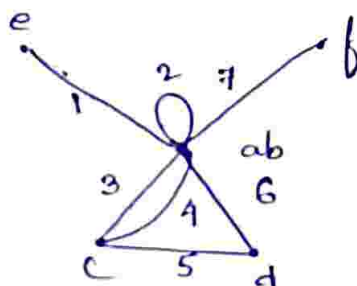
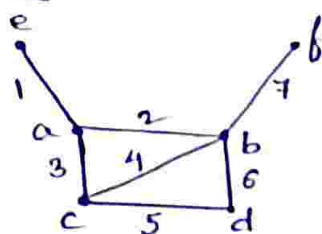


Deletion of edge is like $G-d$.

Deletion of vertex is like $G-v_1$.
(deleting vertex along with its incident edges).

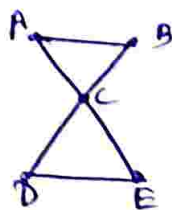


- Fusion:

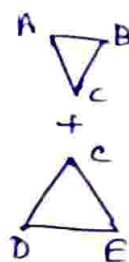


Merger of two vertices.

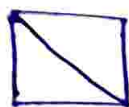
- Theorem: A connected graph is an Euler graph iff it can be decomposed into circuits.



Edge disjoint sub-graphs,

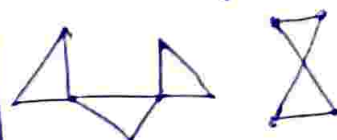
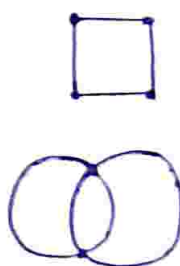


} Form circuits.



Open Euler line,
not an Euler graph
hence doesn't form
edge disjoint circuit.

- Arbitrarily traceable graphs:



Not arbitrarily
traceable as many
possibilities to trace
path that might not
cover all vertex.



This graph is arbitrarily traceable ~~fr~~ from node A.

A graph is arbitrarily traceable if the vertex is present in every circuit.

Computer Architecture: Computer Systems:-

02/02/2023.
NAND (4T)
NOR (4T)
NOT (2T)

- Performance
 - Technology (Base-level) - Transistors
 - Architecture - CISC and RISC

~~NAND~~ AND $\Rightarrow 4T + 2T = 6T$

OR $\Rightarrow 4T + 2T = 6T$

- No. of Transistor per unit Area [increases by Moore's law]

- Illustration of law of diminishing returns:

- Let, $F = 0.25$

- The table shows the speedup $(= 1/(1-F+F/S))$ for various values of S .

S	Speed Up
1	1.00
2	1.14
5	1.25
10	1.29

S	Speedup
50	1.32
100	1.33
1000	1.33
100000	1.33

$$1/(1-0.25) = 1.33$$

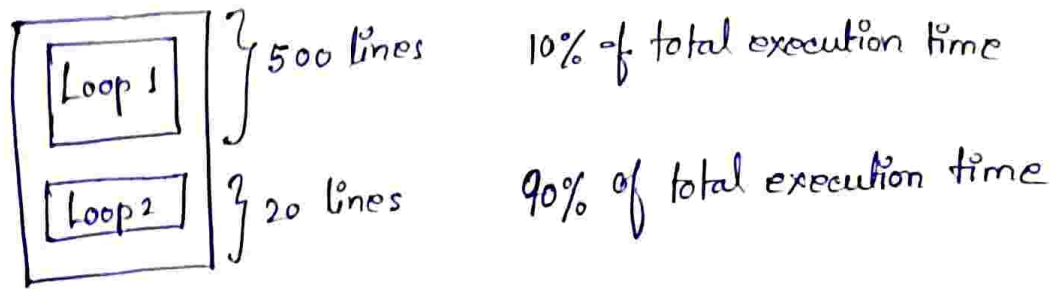
- Let, $F = 0.75$

S	Speed Up
1	1.00
2	1.60
5	2.50
10	3.08

S	Speedup
50	3.77
100	3.88
1000	3.99
100,000	4.00

$$1/(1-0.75) = 4.00$$

- Design Alternative using Amdahl's law:



- Some examples:

→ We make 10% of a program 90X faster, speedup = $1/(0.9 + 0.1/90) = 1.11$ (slowest)
 — " " 90% " " " 10X " " = $1/(0.1 + 0.9/10) = 5.25$
 — " " 25% " " " 25X " " = $1/(0.75 + 0.25/25) = 1.32$
 — " " 50% " " " 50X " " = $1/(0.5 + 0.5/50) = 1.90$
 — " " 90% " " " 50X " " = $1/(0.1 + 0.9/50) = 8.47$ (fastest)

Examples 1

Suppose we are running a set of programs on a RISC processor, for which the following instruction mix is observed:

Operations	Frequency	CPI	$W_i * CPI_i$	% Time	CPI = 2.08
LDA Load	20%	5	1.00	0.48	→ 1/2.08
STA Store	8%	3	0.24	0.12	
Accumulator ALU	60%	1	0.60	0.29	
JUMP Branch	12%	2	0.24	0.11	

We carry out a design enhancement by which the CPI of load instructions reduces from 5 to 2. What will be the overall performance improvement.

$$1/(0.8 +$$

• Example-2

The execution time of a program on a machine is found to be 50 seconds, out of which 42 seconds is consumed by multiply operations. It is required to make the program run 8 times faster. By how much must the speed of the multiplier be improved?

• Example-3:

Suppose we plan to upgrade the processor of a web-server. The CPU is 30 times faster on search queries than the old processor is busy with search queries 80% of the time. Estimate the speedup obtained by the upgrade.

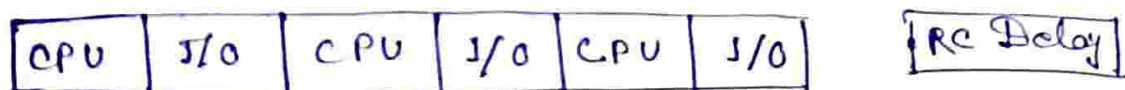
• Example 1:

The total execution time of a typical program is made up of 60% of CPU time and 40% of I/O times. Which of the following alternatives is better?

Ⓐ Increase the CPU speed by 50%.

Ⓑ Reduce the I/O time by half.

Assume that there is no overlap between CPU and I/O operation.



• Amadahl's law: Corollary-1:

Make common case fast.

— Here "common" means most time consuming, and not "most frequent".

— According to Amadahl's law, improving the "uncommon" case will not result in much improvement.

• Amadahl's law Corollary-2:

• Amadahl's law for latency (L)

- By definition, $L_{new} = L_{old} / \text{Speedup}$

- By Amadahl's law, $L_{new} = L_{old} * ((1-F) + F/S)$

- We can write

$$L_{new} = L_{old} * F/S + L_{old} * (1-F)$$

• Extension to Multiple Enhancements:-

Suppose we carry out multiple optimizations to a program:

- Optimization 1 speeds up a fraction F_1 of the program by a factor S_1

- Optimization 2 speeds up a fraction F_2 of the program by a factor S_2

$1 - F_1 - F_2$	F_1	F_2
-----------------	-------	-------

$1 - F_1 - F_2$	F_1/S_1	F_2/S_2
-----------------	-----------	-----------

$$\begin{array}{r} 40 \\ -16 \\ \hline 24 \end{array}$$

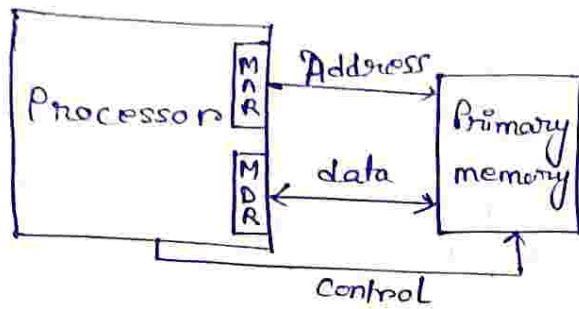
$$\text{SpeedUp} = \frac{1}{(1 - F_1 - F_2) + F_1/S_1 + F_2/S_2}$$

• General expression:

- Assumes m enhancements of fractions $F_1, F_2, \dots, F_m$ by factors of $S_1, S_2, \dots, S_m$ respectively.

$$\text{SpeedUp} = \frac{1}{\left(1 - \sum_{i=1}^m F_i\right) + \sum_{i=1}^m \frac{F_i}{S_i}}$$

Connection between Processor and Memory:



Answer 9: 1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30. 31. 32. 33. 34. 35. 36. 37. 38. 39. 40. 41. 42. 43. 44. 45. 46. 47. 48. 49. 50. 51. 52. 53. 54. 55. 56. 57. 58. 59. 60. 61. 62. 63. 64. 65. 66. 67. 68. 69. 70. 71. 72. 73. 74. 75. 76. 77. 78. 79. 80. 81. 82. 83. 84. 85. 86. 87. 88. 89. 90. 91. 92. 93. 94. 95. 96. 97. 98. 99. 100. 101. 102. 103. 104. 105. 106. 107. 108. 109. 110. 111. 112. 113. 114. 115. 116. 117. 118. 119. 120. 121. 122. 123. 124. 125. 126. 127. 128. 129. 130. 131. 132. 133. 134. 135. 136. 137. 138. 139. 140. 141. 142. 143. 144. 145. 146. 147. 148. 149. 150. 151. 152. 153. 154. 155. 156. 157. 158. 159. 160. 161. 162. 163. 164. 165. 166. 167. 168. 169. 170. 171. 172. 173. 174. 175. 176. 177. 178. 179. 180. 181. 182. 183. 184. 185. 186. 187. 188. 189. 190. 191. 192. 193. 194. 195. 196. 197. 198. 199. 200. 201. 202. 203. 204. 205. 206. 207. 208. 209. 210. 211. 212. 213. 214. 215. 216. 217. 218. 219. 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820. 821. 822. 823. 824. 825. 826. 827. 828. 829. 830. 831. 832. 833. 834. 835. 836. 837. 838. 839. 840. 841. 842. 843. 844. 845. 846. 847. 848. 849. 850. 851. 852. 853. 854. 855. 856. 857. 858. 859. 860. 861. 862. 863. 864. 865. 866. 867. 868. 869. 870. 871. 872. 873. 874. 875. 876. 877. 878. 879. 880. 881. 882. 883. 884. 885. 886. 887. 888. 889. 890. 891. 892. 893. 894. 895. 896. 897. 898. 899. 900. 901. 902. 903. 904. 905. 906. 907. 908. 909. 910. 911. 912. 913. 914. 915. 916. 917. 918. 919. 920. 921. 922. 923. 924. 925. 926. 927. 928. 929. 930. 931. 932. 933. 934. 935. 936. 937. 938. 939. 940. 941. 942. 943. 944. 945. 946. 947. 948. 949. 950. 951. 952. 953. 954. 955. 956. 957. 958. 959. 960. 961. 962. 963. 964. 965. 966. 967. 968. 969. 970. 971. 972. 973. 974. 975. 976. 977. 978. 979. 980. 981. 982. 983. 984. 985. 986. 987. 988. 989. 990. 991. 992. 993. 994. 995. 996. 997. 998. 999. 1000. 1001. 1002. 1003. 1004. 1005. 1006. 1007. 1008. 1009. 1010. 1011. 1012. 1013. 1014. 1015. 1016. 1017. 1018. 1019. 1020. 1021. 1022. 1023. 1024. 1025. 1026. 1027. 1028. 1029. 1030. 1031. 1032. 1033. 1034. 1035. 1036. 1037. 1038. 1039. 1040. 1041. 1042. 1043. 1044. 1045. 1046. 1047. 1048. 1049. 1050. 1051. 1052. 1053. 1054. 1055. 1056. 1057. 1058. 1059. 1060. 1061. 1062. 1063. 1064. 1065. 1066. 1067. 1068. 1069. 1070. 1071. 1072. 1073. 1074. 1075. 1076. 1077. 1078. 1079. 1080. 1081. 1082. 1083. 1084. 1085. 1086. 1087. 1088. 1089. 1090. 1091. 1092. 1093. 1094. 1095. 1096. 1097. 1098. 1099. 1100. 1101. 1102. 1103. 1104. 1105. 1106. 1107. 1108. 1109. 1110. 1111. 1112. 1113. 1114. 1115. 1116. 1117. 1118. 1119. 1120. 1121. 1122. 1123. 1124. 1125. 1126. 1127. 1128. 1129. 1130. 1131. 1132. 1133. 1134. 1135. 1136. 1137. 1138. 1139. 1140. 1141. 1142. 1143. 1144. 1145. 1146. 1147. 1148. 1149. 1150. 1151. 1152. 1153. 1154. 1155. 1156. 1157. 1158. 1159. 1160. 1161. 1162. 1163. 1164. 1165. 1166. 1167. 1168. 1169. 1170. 1171. 1172. 1173. 1174. 1175. 1176. 1177. 1178. 1179. 1180. 1181. 1182. 1183. 1184. 1185. 1186. 1187. 1188. 1189. 1190. 1191. 1192. 1193. 1194. 1195. 1196. 1197. 1198. 1199. 1200. 1201. 1202. 1203. 1204. 1205. 1206. 1207. 1208. 1209. 1210. 1211. 1212. 1213. 1214. 1215. 1216. 1217. 1218. 1219. 1220. 1221. 1222. 1223. 1224. 1225. 1226. 1227. 1228. 1229. 1230. 1231. 1232. 1233. 1234. 1235. 1236. 1237. 1238. 1239. 1240. 1241. 1242. 1243. 1244. 1245. 1246. 1247. 1248. 1249. 1250. 1251. 1252. 1253. 1254. 1255. 1256. 1257. 1258. 1259. 1260. 1261. 1262. 1263. 1264. 1265. 1266. 1267. 1268. 1269. 1270. 1271. 1272. 1273. 1274. 1275. 1276. 1277. 1278. 1279. 1280. 1281. 1282. 1283. 1284. 1285. 1286. 1287. 1288. 1289. 1290. 1291. 1292. 1293. 1294. 1295. 1296. 1297. 1298. 1299. 1300. 1301. 1302. 1303. 1304. 1305. 1306. 1307. 1308. 1309. 1310. 1311. 1312. 1313. 1314. 1315. 1316. 1317. 1318. 1319. 1320. 1321. 1322. 1323. 1324. 1325. 1326. 1327. 1328. 1329. 1330. 1331. 1332. 1333. 1334. 1335. 1336. 1337. 1338. 1339. 1340. 1341. 1342. 1343. 1344. 1345. 1346. 1347. 1348. 1349. 1350. 1351. 1352. 1353. 1354. 1355. 1356. 1357. 1358. 1359. 1360. 1361. 1362. 1363. 1364. 1365. 1366. 1367. 1368. 1369. 1370. 1371. 1372. 1373. 1374. 1375. 1376. 1377. 1378. 1379. 1380. 1381. 1382. 1383. 1384. 1385. 1386. 1387. 1388. 1389. 1390. 1391. 1392. 1393. 1394. 1395. 1396. 1397. 1398. 1399. 1400. 1401. 1402. 1403. 1404. 1405. 1406. 1407. 1408. 1409. 1410. 1411. 1412. 1413. 1414. 1415. 1416. 1417. 1418. 1419. 1420. 1421. 1422. 1423. 1424. 1425. 1426. 1427. 1428. 1429. 1430. 1431. 1432. 1433. 1434. 1435. 1436. 1437. 1438. 1439. 1440. 1441. 1442. 1443. 1444. 1445. 1446. 1447. 1448. 1449. 1450. 1451. 1452. 1453. 1454. 1455. 1456. 1457. 1458. 1459. 1460. 1461. 1462. 1463. 1464. 1465. 1466. 1467. 1468. 1469. 1470. 1471. 1472. 1473. 1474. 1475. 1476. 1477. 1478. 1479. 1480. 1481. 1482. 1483. 1484. 1485. 1486. 1487. 1488. 1489. 1490. 1491. 1492. 1493. 1494. 1495. 1496. 1497. 1498. 1499. 1500. 1501. 1502. 1503. 1504. 1505. 1506. 1507. 1508. 1509. 1510. 1511. 1512. 1513. 1514. 1515. 1516. 1517. 1518. 1519. 1520. 1521. 1522. 1523. 1524. 1525. 1526. 1527. 1528. 1529. 1530. 1531. 1532. 1533. 1534. 1535. 1536. 1537. 1538. 1539. 1540. 1541. 1542. 1543. 1544. 1545. 1546. 1547. 1548. 1549. 1550. 1551. 1552. 1553. 1554. 1555. 1556. 1557. 1558. 1559. 1560. 1561. 1562. 1563. 1564. 1565. 1566. 1567. 1568. 1569. 1570. 1571. 1572. 1573. 1574. 1575. 1576. 1577. 1578. 1579. 1580. 1581. 1582. 1583. 1584. 1585. 1586. 1587. 1588. 1589. 1590. 1591. 1592. 1593. 1594. 1595. 1596. 1597. 1598. 1599. 1600. 1601. 1602. 1603. 1604. 1605. 1606. 1607. 1608. 1609. 1610. 1611. 1612. 1613. 1614. 1615. 1616. 1617. 1618. 1619. 1620. 1621. 1622. 1623. 1624. 1625. 1626. 1627. 1628. 1629. 1630. 1631. 1632. 1633. 1634. 1635. 1636. 1637. 1638. 1639. 1640. 1641. 1642. 1643. 1644. 1645. 1646. 1647. 1648. 1649. 1650. 1651. 1652. 1653. 1654. 1655. 1656. 1657. 1658. 1659. 1660. 1661. 1662. 1663. 1664. 1665. 1666. 1667. 1668. 1669. 1670. 1671. 1672. 1673. 1674. 1675. 1676. 1677. 1678. 1679. 1680. 1681. 1682. 1683. 1684. 1685. 1686. 1687. 1688. 1689. 1690. 1691. 1692. 1693. 1694. 1695. 1696. 1697. 1698. 1699. 1700. 1701. 1702. 1703. 1704. 1705. 1706. 1707. 1708. 1709. 1710. 1711. 1712. 1713. 1714. 1715. 1716. 1717. 1718. 1719. 1720. 1721. 1722. 1723. 1724. 1725. 1726. 1727. 1728. 1729. 1730. 1731. 1732. 1733. 1734. 1735. 1736. 1737. 1738. 1739. 1740. 1741. 1742. 1743. 1744. 1745. 1746. 1747. 1748. 1749. 1750. 1751. 1752. 1753. 1754. 1755. 1756. 1757. 1758. 1759. 1760. 1761. 1762. 1763. 1764. 1765. 1766. 1767. 1768. 1769. 1770. 1771. 1772. 1773. 1774. 1775. 1776. 1777. 1778. 1779. 1780. 1781. 1782. 1783. 1784. 1785. 1786. 1787. 1788. 1789. 1790. 1791. 1792. 1793. 1794. 1795. 1796. 1797. 1798. 1799. 1800. 1801. 1802. 1803. 1804. 1805. 1806. 1807. 1808. 1809. 1810. 1811. 1812. 1813. 1814. 1815. 1816. 1817. 1818. 1819. 1820. 1821. 1822. 1823. 1824. 1825. 1826. 1827. 1828. 1829. 1830. 1831. 1832. 1833. 1834. 1835. 1836. 1837. 1838. 1839. 1840. 1841. 1842. 1843. 1844. 1845. 1846. 1847. 1848. 1849. 1850. 1851. 1852. 1853. 1854. 1855. 1856. 1857. 1858. 1859. 1860. 1861. 1862. 1863. 1864. 1865. 1866. 1867. 1868. 1869. 1870. 1871. 1872. 1873. 1874. 1875. 1876. 1877. 1878. 1879. 1880. 1881. 1882. 1883. 1884. 1885. 1886. 1887. 1888. 1889. 1890. 1891. 1892. 1893. 1894. 1895. 1896. 1897. 1898. 1899. 1900. 1901. 1902. 1903. 1904. 1905. 1906. 1907. 1908. 1909. 1910. 1911. 1912. 1913. 1914. 1915. 1916. 1917. 1918. 1919. 1920. 1921. 1922. 1923. 1924. 1925. 1926. 1927. 1928. 1929. 1930. 1931. 1932. 1933. 1934. 1935. 1936. 1937. 1938. 1939. 1940. 1941. 1942. 1943. 1944. 1945. 1946. 1947. 1948. 1949. 1950. 1951. 1952. 1953. 1954. 1955. 1956. 1957. 1958. 1959. 1960. 1961. 1962. 1963. 1964. 1965. 1966. 1967. 1968. 1969. 1970. 1971. 1972. 1973. 1974. 1975. 1976. 1977. 1978. 1979. 1980. 1981. 1982. 1983. 1984. 1985. 1986. 1987. 1988. 1989. 1990. 1991. 1992. 1993. 1994. 1995. 1996. 1997. 1998. 1999. 2000. 2001. 2002. 2003. 2004. 2005. 2006. 2007. 2008. 2009. 2010. 2011. 2012. 2013. 2014. 2015. 2016. 2017. 2018. 2019. 2020. 2021. 2022. 2023. 2024. 2025. 2026. 2027. 2028. 2029. 2030. 2031. 2032. 2033. 2034. 2035. 2036. 2037. 2038. 2039. 2040. 2041. 2042. 2043. 2044. 2045. 2046. 2047. 2048. 2049. 2050. 2051. 2052. 2053. 2054. 2055. 2056. 2057. 2058. 2059. 2060. 2061. 2062. 2063. 2064. 2065. 2066. 2067. 2068. 2069. 2070. 2071. 2072. 2073. 2074. 2075. 2076. 2077. 2078. 2079. 2080. 2081. 2082. 2083. 2084. 2085. 2086. 2087. 2088. 2089. 2090. 2091. 2092. 2093. 2094. 2095. 2096. 2097. 2098. 2099. 2100. 2101. 2102. 2103. 2104. 2105. 2106. 2107. 2108. 2109. 2110. 2111. 2112. 2113. 2114. 2115. 2116. 2117. 2118. 2119. 2120. 2121. 2122. 2123. 2124. 2125. 2126. 2127. 2128. 2129. 2130. 2131. 2132. 2133. 2134. 2135. 2136. 2137. 2138. 2139. 2140. 2141. 2142. 2143. 2144. 2145. 2146. 2147. 2148. 2149. 2150. 2151. 2152. 2153. 2154. 2155. 2156. 2157. 2158. 2159. 2160. 2161. 2162. 2163. 2164. 2165. 2166. 2167. 2168. 2169. 2170. 2171. 2172. 2173. 2174. 2175. 2176. 2177. 2178. 2179. 2180. 2181. 2182. 2183. 2184. 2185. 2186. 2

Computer Architecture:-

06/02/2022.

• Problem faced in Amdahl's law: Cache memory

• Flash memory: Pen-drives

• Types of memory: Registers, RAM, ROM, Hard disk, Magnetic disk, Flash memory, Flip-flops, Cache memory, CMOS (semiconductors) [Logic gates are made], Virtual memory.

• Memory performance parameters:

(i) Speed - Reading and writing speed

(ii) Cost -

(iii) Spa. Storage density -

(iv) Proximity to the ALU.

} Trade-off means if one goes down others might go up.

• Difference between RAM and ROM. [Page-376]

↳ • ROM is where data once stored is permanent or semi-permanent.
e.g. ROM, EPROM, PROM, EEPROM.

• RAM is one where data access time is the same independent of the location (address).

e.g. SRAM: retained as long as power is on.

DRAM: requires periodic refreshing even when power is on.

• Terminologies used to measure speed of the memory system.

(a) Memory Access Time: Time b/w initiation of an operation (Read or write) and completion of that operation.

(b) Latency: Initial delay from the initiation of an operation to the time the first data is available.

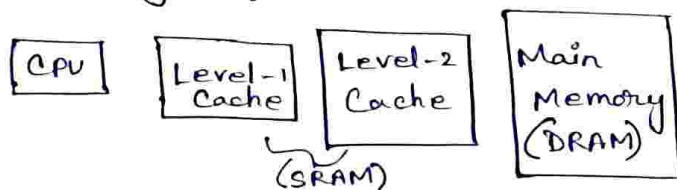
(c) Bandwidth: Maximum speed of data transfer in bytes per second. (in internet connection).

• Reduce Memory access time and latency, increase bandwidth.

- Cache memory: Concept to increase the effective speed of the memory system.
- Virtual memory: Concept to increase the effective size of the memory system.

↓
Main + secondary

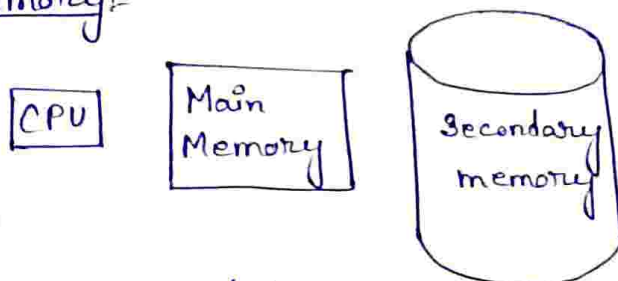
- Cache memory: Page - 381.



SRAM more expensive than DRAM.

- A fast memory (possibly organized in several levels) that sits between processor and main memory.
- Faster than main memory and relatively small.
- Frequently accessed data and instructions are stored here.
- Cache memory makes use of the fast SRAM technology.

- Virtual memory:



- Technique used by the operating system to provide an illusion of very large memory to the processor.
- Program and data are actually stored on secondary memory that is much larger.
- Transfer parts of program and data from secondary memory to main memory only when needed.

• External connection requirements:

→ The 8×4 memory requires the following external connections:

- Address decoder of size: 3×8 .

• 3 external connections for address.

- Data output: 4-bit.

• 4 external connections for data.

- 2 external connections for R/W' and CS'.

- 2 external connections for power supply and ground.

- Total of $3+4+2+2=11$.

• What about a 256×16 Memory?

→ Here the total number of external connection are estimated as follows:

- Address decoder size: 8×256

• 8 external connections for address

- Data output: 16-bit

• 16 external connections for data.

- 2 external connections for R/W' and CS'.

- 2 external connections for power supply and ground.

- Total of $8+16+2+2=28$.

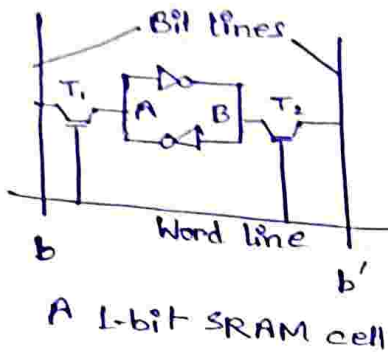
• Broadly two types of semiconductor memory systems:

→ SRAM - 6T (1 unit SRAM) - Cache memory

→ DRAM - [Asynchronous DRAM - 1C-1T (1 unit DRAM)
Synchronous DRAM - Main memory

C = Capacitor and T - Transistor.

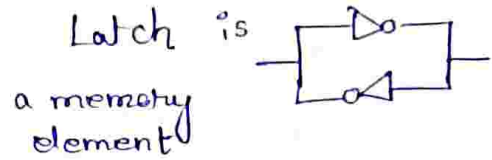
• SRAM - memory arrays can be arranged in rows and columns of memory cells called word line and bit line.



$$2 \text{ inverters} = 2T + 2T$$

$$\therefore \text{SRAM cell} = 2T + 2T + T_1 + T_2 = 6T$$

2T + Latch

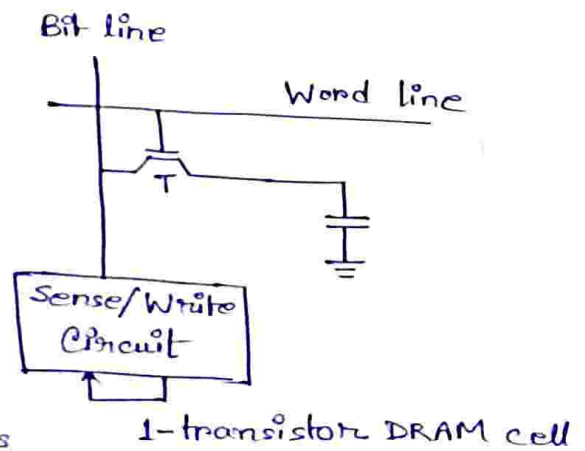


• Features of SRAM:-

- ① Moderate/High power consumption.
- ② Simplicity - refresh circuitry is not needed.
- ③ Fast operation
- ④ High cost
- ⑤ Limited capacity.

• Feature DRAM

- ① Dynamic RAM do not retain its state even if power supply is on.
 - ② Requires periodic refresh.
 - ③ Less expensive than SRAM
 - ④ Address lines are multiplexed.
- refreshing means charging and discharging of capacitors
be 2%



• Memory design hierarchy:-

• Flash-memory:-

- Non-volatile memory technology that uses floating-gates MOS transistors.
- slower than DRAM, but higher packing density and lower cost per bit.

• Magnetic disk:

- Provides large amount of storage, with very low cost per bit.
- Much slower than DRAM, and also flash memory.
- Requires mechanical moving parts, and uses magnetic recording technology.

• Typical hierarchy

- (i) Processor registers
- (ii) Level-1-cache
- (iii) Level-2-Cache
- (iv) Level-3-Cache.
- (v) Main memory
- (vi) Secondary memory.

• As we move away from processor.

Data Structures:

Operations on queue (implemented using arrays).

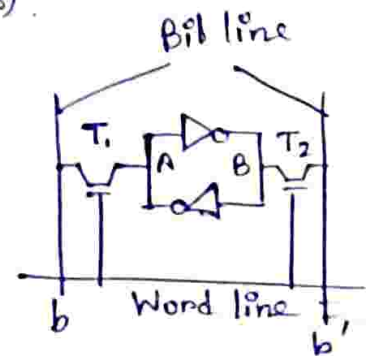
Algorithm enqueue (item, q, n, rear)

{
 // insert

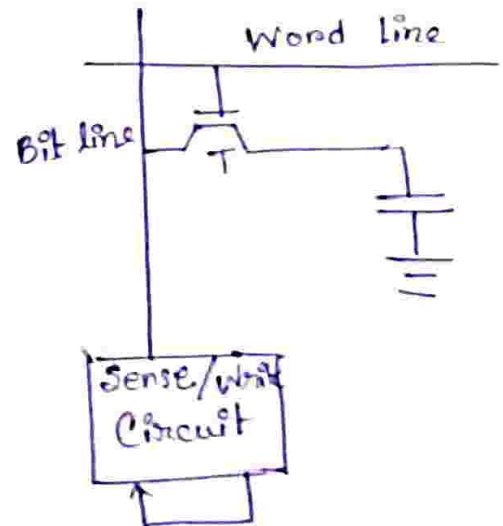
$$\begin{array}{r} \text{Five} \quad 165 \\ - 15 \\ \hline 180 \\ + \end{array}$$

$$\begin{array}{r} 5 \quad 15 \quad 25 \quad 40 \quad 100 \\ 10 \quad 10 \quad 25 \quad 60 \\ 0 \quad 15 \quad 35 \\ \hline 25 \end{array}$$

07/02/2023



1-bit SRAM cell.

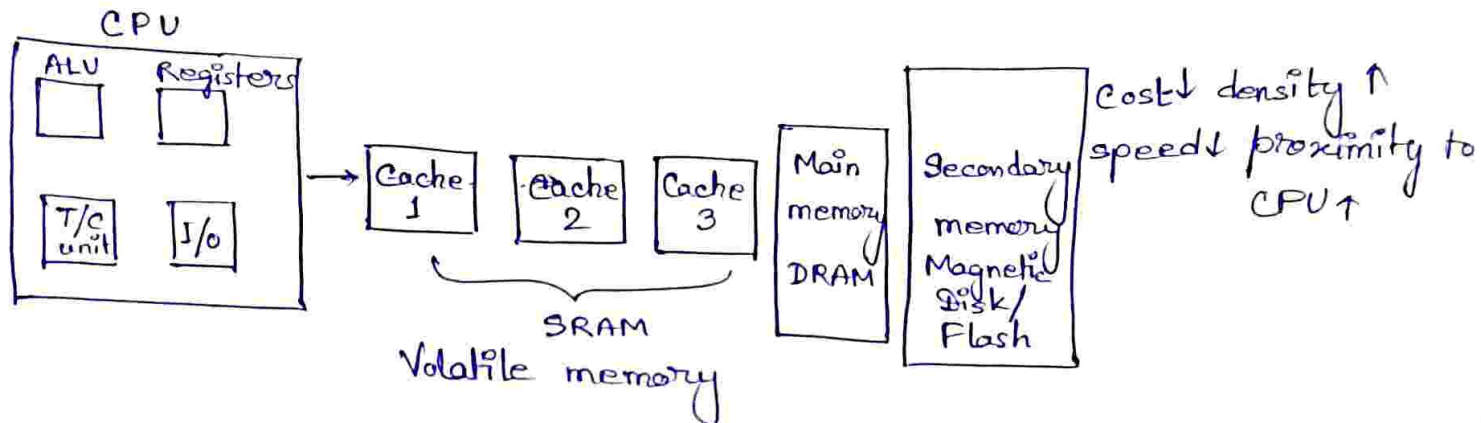


25/2

Computer Architecture:-

08/02/2023

- ① Registers $\begin{cases} \text{GPR} \\ \text{SPR} \end{cases}$
- Faster
 - Costlier than RAM
 - Low storage density (limited capacity)
 - Within CPU.



Ideal memory access time = 1 CPU cycle.

Real memory access time $\gg$ 1 CPU cycle.

$$\text{Minimum CPU stall cycles} = \frac{\text{Memory access time}}{\text{Clock cycle} - 1}$$

It is:

- Memory Latency Reduction Techniques:
 - Faster DRAM cells (depends on VLSI Technology)
 - Wider memory bus width (fewer memory access needed)
 - Multiple memory banks.
 - Integration of memory controller with processor
 - New emerging RAM technologies.
- Memory Latency Hiding Techniques:
 - Memory hierarchy (using SRAM based cache memories)
 - Pre-fetching instructions and/or data from memory before they are actually needed (used to hide long latency)

▪ Locality of Reference:-

- Programs tend to reuse data and instructions they have used recently.
- Rule of thumb: 90% of the ~~time~~ total Execution time of a program is spent in only 10% of the code (also called 90/10 rule).
- Reason: nested loops in a program, few procedures calling each other repeatedly, arrays of data items being accessed sequentially, etc.
- Basic idea to exploit this rule:
—

- The 90/10 rule has two dimensions:
 - a) Temporal Locality (locality in time).
 - b) Spatial Locality (locality in space).

• Performance of Memory Hierarchy:-

→ Hit Ratio/Hit Rate:

- The hit ratio H is defined as the probability that a logical address generated by the CPU refers to information in M_1 .
- We can determine H experimentally as follows:
 - A set of representative programs is executed or simulated
 - The no. of references to M_1 and M_2 , denoted by N_1 and N_2 respectively, are recorded.

$$H = \frac{N_1}{N_1 + N_2}$$

- The quantity $(1-H)$ is called the miss ratio.

• Access Time:

- Let, t_{A1} and t_{A2} denote the access time of M_1 and M_2 respectively, relative to the CPU.

The average time required by the CPU to access a word in memory can be expressed as:

$$t_A = H \cdot t_{A1} + (1-H) \cdot t_{Miss}$$

where t_{Miss} denotes the time required to handle the miss, called miss penalty.

- If t_B denotes the block transfer time, we can write,
 $t_{Miss} = t_B + t_{A1}$ [since $t_B \gg t_{A1}, t_{A2} \approx t_B$]

Thus, $t_A = H \cdot t_{A1} + (1-H)(t_B + t_{A1})$

- If t_{Hit} denotes the time required to check whether there is a hit, we can write $t_{Miss} = t_{Hit} + t_B + t_{A1}$.

• Efficiency:

- Let, $r = \frac{t_{A2}}{t_{A1}}$ denote the access time ratio of the two levels of memory.

- We define the access efficiency as $e = \frac{t_{A1}}{t_A}$, which is the factor by which t_A differs from its minimum possible value.

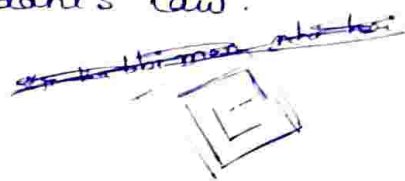
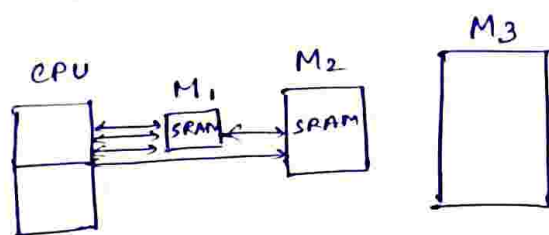
$$\text{Efficiency, } e = \frac{t_{A1}}{H \cdot t_{A1} + (1-H) \cdot t_{A2}} = \frac{1}{H + (1-H) \cdot r}$$

• Speed up:-

- The speed up gained by using the memory hierarchy is defined as $S = \frac{t_{A2}}{t_A}$.

- We can write it as: $S = \frac{t_{A2}}{H \cdot t_{A1} + (1-H) \cdot t_{A2}} = \frac{1}{H/r + (1-H)}$

- The same results follow from Amdahl's Law.



Q. Hit/miss rate, speed up and efficiency for M_1, M_2, M_3, M_4 .

Data Structures:-

09/02/2023

▪ Space complexity/Time complexity:-

- The space complexity of an algorithm is the amount of memory it needs to run to completion.
- The time complexity of an algorithm is the amount of computer time it needs to run to completion.

The space needed by each of the algorithms is seen to be the sum of the following components:

① A fixed part that is independent of the characteristics (eg number size) of the inputs and outputs. This p

$$S(p) = \text{space requirement} = c + S_p(\text{instance characteristics})$$

$p = \text{algorithm}$
 $c = \text{constant}$

② Run time is denoted by $t_p(\text{instance characteristics})$.

COMPUTER ARCHITECTURE:-

17/02/2023.

Some common Terminologies:-

- **Block:** The smallest unit of information transferred between two levels.
- **Hit rate:** The fraction of memory accesses found in the upper level.
- **Hit time:** Time to access the upper level.
 - Upper Level access time + Time to determine hit/miss.
- **Miss:** Data item needs to be retrieved from a block in the lower level.
- **Miss Rate:** The fraction of memory accesses not found in the upper level.
- **Miss Penalty:** Overhead whenever a miss occurs.
 - Time to replace a block in the upper level + Time to transfer the missed block.

Q. How is a block found if present in cache?

- Page - 492 - Slides.
- **Use of a valid bit:**
 - There must be a way to know whether a cache block contains valid or garbage information.
 - A valid bit can be added to the TAG, which indicates whether the block contains valid data.

- If the valid bit is not set, there is no need to match the corresponding TAG.

Q. 2. Which block should be replaced on a cache miss?

↳ Two primary strategies are used:

- a) Random: The candidate block is selected randomly for replacement. The simple strategy tends to spread allocation uniformly.
- b) Least Recently Used (LRU): The block replaced is the one that has not been used for the longest period of time.

• Types of Cache Misses:-

1. Compulsory Miss:-

- On the first access to a block, the block must be brought into the cache.
- Also known as cold start misses, or first reference miss.
- Can be reduced by increasing cache block size or prefetching cache blocks

2. Capacity Miss:-

- Blocks may be replaced from cache because the cache cannot hold all the blocks needed by a program.
- Can be reduced by increasing the total cache-size.

3. Conflict Miss:-

- In case of direct mapping or N-way set associative mapping, several blocks may be mapped to the same block or set in the cache.
- May result in block replacements and hence access misses, even though all the cache blocks may not be occupied.
- Can be reduced by increasing the value of N (cache associativity).

Q.4. What happens on a write?

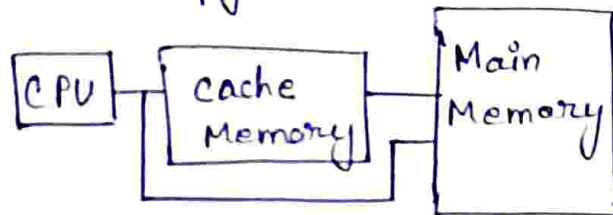
→ Page - 499 - Slides.

Latches
Flip-flops
Buffers
Registers

• Cache write Strategies.

→ Cache designs can be classified based on the write and memory update strategy being used.

1. Write Through / Store Through
2. Write Back / Copy Back.



a) Write Through Strategy: Page - 501

— A write buffer is often used to reduce CPU write stall time while data is written to main memory.

b) Write Back Strategy: Page - 503

- Write-back cache blocks can be clean or dirty.

—

• Cache Write Miss Policy: Page - 505.

- Write Allocate:

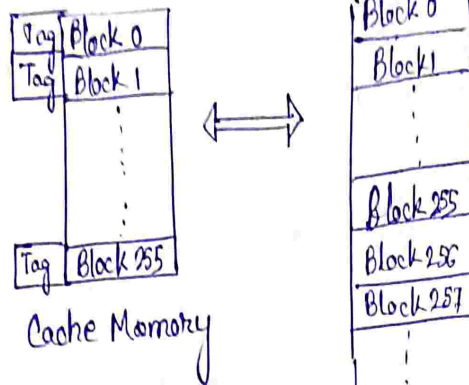
- No-Write Allocate:

• Typical usage: Page - 506

→ Write-back cache with write-allocate.

→ Write-through cache ~~with no-write~~ with no-write allocate.

Direct Mapping



Tag	Block	Word
11	8	5

Memory Address

Fully Associative Mapping

Tag	Word
19	5

Memory Address

4-way Set Associative Mapping

Tag	Set	Word
13	6	5

Memory Address

Set 0
Set 1
...

Set 63
